

MAML-Select: An Online Adaptive Client Selection Method for Federated Learning via Meta-Learning

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We sincerely thank the Editor, the Associate Editor, and the anonymous Reviewer for their valuable time and effort in evaluating our manuscript. The comments were detailed and constructive, and they have helped us improve the technical accuracy and the clarity of our work. In the revised version, we have carefully addressed every raised concern. *All changes made in response to the Reviewer's comments have been clearly highlighted in blue, as per the journal's guidelines.* Below, we provide a detailed point-by-point response to each comment. Please note that the Reviewer's comments are *italicized*, followed by our responses in normal font. Along with this letter we submit (a) a clean version of the revised manuscript and (b) a coloured marked version of the revised manuscript, in which the changes are highlighted.

This revision does not include a supplementary file. Every item the Reviewer raises, and all content that previously appeared only in the supplementary material, is now in the main manuscript. The method is therefore described in exactly one place, and that description was written from the selector source code. The implementation is also released publicly, so the description can be checked against the code that produced the results.

Response to Comments of Reviewer 1

Comment 1: *The main manuscript and the supplementary material describe different optimization procedures for the same method.*

Reply 1: We thank the Reviewer for identifying this. We checked both descriptions against the selector source and corrected the manuscript to match the code on five points. The outer meta-update acts on the query set. Both meta-updates occur at the start of the round, before any client is scored. The support set is the cohort of round $t - 2$ and the query set is the cohort of round $t - 1$, so $\mathcal{D}_{sup}(t) = \mathcal{D}_{query}(t - 1)$. The exploration step is deterministic and gives one slot to the available client of largest staleness, with the lowest selection frequency breaking ties. The six state features are standardized across the available pool every round.

The lag between the two sets is what makes the procedure a meta-learning problem rather than online regression. The inner step adapts on an older round and the meta-gradient is measured on a newer one, so the outer update rewards initializations from which one adaptation step transfers forward in time. Please refer to Eq. (4) for the standardization, Eq. (5) for the lag, the opening of Section IV-C for the phase ordering, and

Algorithm 1 of the revised manuscript.

Comment 2: *The latency-cost function is defined inconsistently between the two documents.*

Reply 2: We thank the Reviewer for this correction. The implementation normalizes the deadline overrun by the target latency,

$$c_{i,t} = \lambda \frac{(T_{i,t} - T_{target})_+}{T_{target}} - \Delta_{i,t},$$

and every reported number was produced with this form. The submitted equation omitted the division by T_{target} , and the revised manuscript uses the normalized form throughout.

The normalization also carries meaning. It makes the deadline overrun a dimensionless fraction, so both terms of the cost are on the same scale and a single value of λ transfers across datasets with different round times. We have also removed the earlier ambiguity in T_{target} , which is the mean expected round latency of the Tier-2 devices in the available pool, recomputed each round, with the pool median as a fallback when no Tier-2 device is available.

Checking the latency definitions against the simulator in the same pass showed that the submitted computation time carried a per-sample model cost the simulator does not apply, and that the energy accumulates over the whole per-round latency rather than over the computation time alone. Both definitions now match the code. Please refer to Eq. (5) for the computation time, Eq. (7) for the cost, and Eq. (13) for the energy of the revised manuscript.

Comment 3: *The per-client ranking score is not derived from the cohort-level objective it is supposed to serve.*

Reply 3: We agree, and the revised manuscript now gives the derivation in two steps and states which step is exact.

The first step is exact. The straggler term is a maximum and is therefore not separable, but for any $T_{target} > 0$ it is bounded by $1 + \sum_{i \in \mathcal{S}_t} \rho_{i,t}$ with $\rho_{i,t} = (T_{i,t} - T_{target})_+ / T_{target}$. The second step is the one approximation. The risk decrease is unavailable at selection time, so we substitute the observable local loss reductions. This treats cohort contributions as additive and equates local with global progress, and the manuscript now says so.

The result is a modular cohort surrogate, and Proposition 1 proves by an exchange argument that Top- K is an exact minimizer of it under the cardinality constraint. Because the costs are predicted rather than realized, we claim that the cohort exactly optimizes a stated surrogate under predicted costs, and not that it optimizes the original objective. Please refer to the subsection *From the Cohort Objective to a Per-Client Score* in Section IV and to Proposition 1 of the revised manuscript.

Comment 4: *Because the supplementary reveals a forced client-coverage stage, the fairness and coverage results must be re-evaluated.*

Reply 4: We agree that this had to be answered with data. We discarded the forced cold-start rounds and

recomputed both metrics over the policy-driven rounds alone, across all 74 distinct MAML-Select runs in the study and every λ , inner-step, width and feature-ablation setting.

Dataset	Runs	Cov. (%) all	Cov. (%) post-start	Jain all	Jain post-start
Fashion-MNIST	48	100	100	0.755	0.737
CIFAR-10	22	100	100	0.411	0.367
CIFAR-100	4	100	100	0.802	0.785

Coverage is 100% with and without the cold start, and this holds in every individual run rather than on average. Jain’s index falls by at most 0.044. Proposition 2 shows why this is not a coincidence. One slot per round goes to the client of maximal staleness, and a client that takes that slot has its staleness reset, so each of the other $N - 1$ clients can block a given client at most once and every client is selected within any window of $N - 1$ consecutive rounds. At $N = 100$ and $T = 200$ the condition is met twice over.

The revised manuscript also separates the two metrics rather than presenting them as equivalent evidence. Coverage is a structural property guaranteed by the exploration slot. Jain’s index measures how evenly the remaining places are distributed, is genuinely lower than for the uniform selectors, and reflects a latency-aware selector preferring faster devices. Both full-run and post-cold-start values are reported. Please refer to Section V-C and Proposition 2 of the revised manuscript.

Comment 5: *The CriticalFL coverage values are contradictory between the main table and the supplementary table.*

Reply 5: The Reviewer is right. The two values cannot both be true, because Jain’s index over N clients satisfies $J \leq \text{Cov}$, so a Jain index of 0.98 is impossible at 33% coverage. We now use this invariant as a check on the reported table.

The cause was a reporting-script error rather than an experimental one. The script that produced the supplementary table read a progress file that is written continuously during a run, and for the CriticalFL runs it read snapshots taken while those runs were still in progress. It therefore reported partial-round coverage against a final-round Jain index. The main-paper value of 100% is the correct one.

We recomputed from the completed per-round logs rather than from any progress file. On CIFAR-100, where the full baseline sweep is available, CriticalFL reaches 100% coverage with $J = 0.976$, which is consistent with the main table and with the invariant above. The CIFAR-100 block of Table II and the tier-participation figures are computed on the same basis. Withdrawing the supplementary material removes the contradiction structurally, since there is now one table with one set of values. Please refer to Table II and Section V-B of the revised manuscript.

Comment 6: *The approximation-error and convergence claims need strengthening, since the theory assumes a stationary objective while the method is motivated by non-stationary client utility.*

Reply 6: We agree. The submitted corollary assumed a stationary query objective, and that assumption is untenable here. Because $\mathcal{D}_{sup}(t) = \mathcal{D}_{query}(t - 1)$, assuming it collapses the inner and outer objectives into one and dissolves the meta-learning problem into descent on a fixed loss. It also asserts that client utility does not change during training, which is the premise the method is built on. Over 199 logged rounds the realized query objective is non-monotone on all three datasets, with round-to-round changes reaching 2.67, 0.70 and 3.61.

We have replaced that statement with a tracking bound that carries the drift explicitly. With $V_T = \sum_t (q_{t+1}(\phi_{t+1}) - q_t(\phi_{t+1}))$ measured at a common iterate, Corollary 1 gives

$$\min_{1 \leq t \leq T} \|\nabla q_t(\phi_t)\|_2^2 \leq \frac{2(q_1(\phi_1) - q_{\min})}{\eta T} + \frac{2V_T}{\eta T} + \frac{1}{T} \sum_{t=1}^T \delta_t^2.$$

The proof is the same telescoping argument, except that consecutive function values belong to different objectives, so the drift enters explicitly instead of being assumed away.

We have since measured V_T rather than only arguing for it. Recording the two consecutive objectives at a common iterate gives $V_T = -2.89$ on Fashion-MNIST, 0.51 on CIFAR-10 and 1.79 on CIFAR-100 over 198 rounds. The partial sums flatten rather than growing with the horizon, so the measured drift lies in the $V_T = o(T)$ regime, and the Fashion-MNIST value is negative and therefore in the benign case. These are single instrumented runs, one per dataset, so we present them as a measurement of the drift term and not as a distributional claim.

Two further points follow from the Reviewer’s comment. The quantity δ_t was previously called the first-order approximation error. It is not the term dropped by the first-order approximation but the displacement of the applied direction from the query gradient at the un-adapted weights, and the revised text says so. The lemmas and the corollary also analyse a plain-descent outer step, whereas the implementation passes that direction through Adam, so the analysis characterizes the descent direction but not Adam’s per-coordinate rescaling. Remark 1 states both points and records the extension to adaptive-moment methods under a drifting objective as open.

Section V-D reports the empirical checks. The inner step is non-increasing on the support loss in 198 of 198 adapted rounds on all three datasets, and the meta-update magnitude settles at a small non-zero level rather than decaying to zero, which is the behaviour the tracking bound predicts. Please refer to Corollary 1, Remark 1 and Section V-D of the revised manuscript.

We note one point of experimental procedure. The CriticalFL runs on CIFAR-100 terminated at rounds 181 and 150 of the planned 200, so every CIFAR-100 comparison is made at the matched horizon of round 150. This applies uniformly to all methods in that block.

We thank the Reviewer once again for the careful reading, which has improved the accuracy of the method description, the derivation and the theory.